1. Scanning Image vs OCR Text

Documents can be shared either by scanning and emailing the image or by using OCR and sending a text file. Scanning an image keeps the document exactly as it is, including handwriting, signatures, tables, diagrams, and formatting. It does not need any extra software or correction. This method is best for handwritten notes, signed documents, or complex layouts. However, scanned images have large file sizes, take more time to send, and the text cannot be searched, copied, or edited. Screen readers also cannot read image-based text.

Using OCR converts the document into text form. Text files are small in size and easy to send. The text can be edited, searched, copied, and stored. OCR text is accessible to screen readers and useful when the content needs to be analyzed or reused. The disadvantage is that OCR may make mistakes if the handwriting is unclear or the image quality is poor. Tables and diagrams may not be preserved properly.

Scanning is preferred for handwritten or visually complex documents, while OCR is preferred for typed, editable, and searchable documents.

2. Spam Detection in Emails

Spam emails usually contain suspicious words such as “free,” “win now,” “urgent,” and “act immediately.” They may promise unrealistic rewards or prizes and often come from unknown or fake senders. These emails may ask for personal details like passwords or bank information. Poor grammar, spelling errors, excessive capitalization, and generic greetings such as “Dear customer” are also common signs.

Spam filters analyze word frequency, subject line patterns, formatting styles, character patterns like deliberate misspellings, and sentence structure. After detection, emails can be moved to the spam folder, labeled if confidence is low, or deleted only when detection confidence is very high to avoid loss of important emails.

3. Autonomous Vehicle System

An autonomous vehicle must follow traffic rules, speed limits, and local transport regulations. It should consider road conditions, weather, braking distance, turning radius, obstacles, passenger comfort, and fuel or battery limits. Inputs to the system include sensor data from LIDAR, radar, cameras, GPS, and ultrasonic sensors, along with traffic signals, road signs, live traffic data, passenger instructions, map data, and vehicle status.

The system outputs include steering control, acceleration, braking, indicators, navigation commands, and communication messages. Information is shared with passengers through displays, audio alerts, and mobile applications. Vehicle-to-vehicle communication is important to prevent accidents and improve traffic flow.

4. Circle and Ellipse Fitting

For a circle model, the parameters are the center (h, k) and radius r. The objective is to minimize the error between the data points and the circle equation. An ellipse is a more general model with five parameters: center (h, k), semi-major axis a, semi-minor axis b, and orientation angle θ. An ellipse is preferred because real-world data often shows unequal spread and different variations in different directions. A circle is a special case of an ellipse when a equals b.

iv) A circle is a special case of an ellipse when a=b.

6. Yes, median can be calculated for this data

Given the data: 75, 63, 100+, 36, 51, 45, 80, 90

After arranging in ascending order: 36, 45, 51, 63, 75, 80, 90, 100+

The median is calculated as:

Median = (63 + 75) / 2 = 69 hours